

Image Segmentation: TransUNet vs. UNet++ on ISIC 2018

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https://github.com/claimiris/image_segmentation

ABSTRACT

Segmentation of skin lesions constitutes the foundational step in computer-assisted diagnosis for melanoma and other skin diseases. In this study, we conducted a comparison between two recent deep neural network architectures, namely TransUNet and UNet++. TransUNet consists of a CNN encoder based on the ResNet architecture, supplemented with a transformer-based bottleneck to learn spatial relationships among pixels, whereas UNet++ adopts an encoder-decoder architecture, which is further supported by dense skip connections through the incorporation of additional sub-networks in the decoder segment. Training was conducted for 10 epochs using a hybrid **Dice-Focal loss** function with **AdamW** optimizer and warm-up cosine learning rate decay. On the held-out test data TransUNet achieves a Dice score of **0.828** and IoU Score of **0.737**, whereas UNet++ gave a Dice of **0.813** and IoU of **0.719**, along with a smaller Hausdorff distance of **18.2 px** vs. **20.5 px**. Our analysis highlights complementary strengths: TransUNet shows stronger performance in terms of global semantic coherence, while UNet++ results in a tighter contour around the target.

INTRODUCTION

Skin cancer is one of the most common cancers in the world, with melanoma being the cause of most skin-cancer deaths. While dermoscopy allows visualization of the sub-surface architecture of the skin without the need of invasive procedures, segmentation of the lesion on dermoscopic images is a painstaking task prone to intraobserver variation. Automated segmentation provides a better approach for defining the region of interest that will be used in subsequent classification and grading algorithms

The ISIC 2018 challenge [5] has proposed a benchmarking standard for the segmentation of skin lesions, which has more than **2,500** dermoscopic images along with binary masks labeled by experts. Deep learning-based models, especially the UNet [1] model and its variants, perform exceptionally well on this benchmark. Nonetheless, fully convolutional models cannot adequately capture global context information, an issue that has been addressed in Vision Transformer (ViT) [4]

TransUNet [3] is the architectural implementation that initially incorporated the use of transformer self-attention mechanisms in the UNet encoder-decoder architecture, providing remarkable performance for multi-organ segmentation. On the other hand, there have been alternative solutions like the

UNet++ [2], which adopt a complementary approach in the sense of modifying the skip-connections by using nested networks and deep supervision.

Main Contributions

- In this work we implemented both architectures in PyTorch from scratch, having a unified training loop, loss function, and pipeline for evaluation.
- Trained both architectures for 10 epochs each on ISIC 2018 dataset using the same data splits and augmentations
- Conducted quantitative comparison between the two architectures using five different metrics: Dice, IoU, Sensitivity, Specificity, and HD95
- Compared the qualitative results by looking at segmentation maps of the two architectures to better understand their weaknesses

RELATED WORK

Encoder-Decoder Segmentation Networks

UNet [1] pioneered the classical encoder-decoder with skip connection model for biomedical image segmentation with symmetrical skip connections to preserve fine-scaled information that was lost during pooling operations. In turn, SegNet [7] and DeepLab [8] improved the architecture with pooling memory indices and dilated convolutions respectively. UNet++ [2] model reconstructs skip connections into denser sub-networks for decoder feature aggregation at all resolutions.

Transformer-Based Segmentation

Vision Transformers [4] represent image patches as token sequences and use the mechanism of multi-head self-attention to learn global dependencies by accounting for the small receptive field of convolution operations. TransUNet [3] uses a transformer encoder in the bottleneck part of a hybrid model comprising CNN and transformers; the SETR [9] model uses a transformer-based encoder only. Swin-UNet [10] uses only the Transformer architecture to build an UNet model.

Skin Lesion Segmentation

Early ISIC challenge winners relied on ensembles of hand-crafted features. More recently semi-supervised [13] and attention-gated [12] variants of UNet have achieved strong results. The two-stage approach of segmenting coarse regions

followed by boundary refinement has also proven effective. Our work focuses on a clean single-stage comparison under a unified loss and training regime.

METHODOLOGY

TransUNet Architecture

Our TransUNet implementation follows a three-stage pipeline.

CNN Encoder: A stem convolution (7×7 , stride 2, 64 channels) feeds three Pre-Activation Bottleneck blocks with Group Normalisation and Standardised Convolutions [14], producing feature maps of 64, 512, and 1024 channels at successively halved spatial resolutions ($1/2$, $1/4$, $1/8$ of input).

Transformer Bottleneck: A projection to a 768-dimensional embedding through a 1×1 convolution is made for the 1024 channels feature map. The spatial positions get flattened and then embedded into tokens and further augmented with the positional embeddings. This embedding goes through four Transformer blocks each containing pre-LayerNorm Multi-Head Self-Attention (12 heads) and two layers GELU MLP.

CNN Decode: The reshaped output of the transformer network is fed into two transposed convolution upsamplings steps ($768 \rightarrow 256 \rightarrow 64$ channels, each doubling resolution), then bilinear upsampling to the original size followed by a 1×1 convolution layer outputting a single-channel logit map.

The model uses a total number of about 44M parameters.

UNet++ Architecture

UNet++ extends UNet by introducing intermediate nodes $\mathbf{x}^{i,j}$ at every intersection of encoder depth i and decoder column j . Each node aggregates all preceding same-resolution features:

$$\mathbf{x}^{i,j} = \mathcal{H}\left(\left[\mathbf{x}^{i,k}\right]_{k=0}^{j-1}, \mathcal{U}(\mathbf{x}^{i+1,j-1})\right) \quad (1)$$

where $\mathcal{H}$ is a double ConvBnReLU block and $\mathcal{U}$ is bilinear upsampling followed by a 1×1 projection. Our encoder uses five stages with channel widths [32, 64, 128, 256, 512]. Deep supervision generates auxiliary outputs from columns $j = 1, 2, 3, 4$, weighted at training time with weights [0.1, 0.2, 0.3, 0.4]. At inference, the four outputs are averaged to obtain the final prediction. The total parameter count is approximately 9M.

Loss Function

We use a **Hybrid Dice + Focal loss**:

$$\mathcal{L} = \lambda_{\text{Dice}} \mathcal{L}_{\text{Dice}} + \lambda_{\text{focal}} \mathcal{L}_{\text{focal}} \quad (2)$$

with $\lambda_{\text{Dice}} = \lambda_{\text{focal}} = 0.5$. The Dice loss directly maximises mask overlap:

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2 \sum p_i t_i + \epsilon}{\sum p_i + \sum t_i + \epsilon} \quad (3)$$

The Focal loss [11] addresses foreground–background class imbalance by down-weighting easy negatives:

$$\mathcal{L}_{\text{focal}} = - \sum_i \alpha_i (1 - p_i)^\gamma \log(p_i) \quad (4)$$

with $\alpha = 0.75$ and $\gamma = 2.0$. For UNet++ the hybrid loss is aggregated across all deep-supervision heads via $\mathcal{L}_{\text{DS}} = \sum_j w_j \mathcal{L}^{(j)}$.

Training Configuration

Both models are trained under identical conditions. Images are resized to 256×256 pixels and normalised with ImageNet statistics. The training pipeline uses the following augmentations: horizontal and vertical flipping ($p = 0.5$ each), random 90° rotation ($p = 0.5$), shift–scale–rotate with reflection padding ($p = 0.5$), elastic deformation ($p = 0.3$), and random brightness/contrast jitter ($p = 0.5$).

- **Optimiser:** AdamW, $\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay 10^{-4}
- **Learning rate:** 3×10^{-4} with a 5-epoch linear warmup followed by cosine decay to 10^{-6}
- **Batch size:** 8 (training), 1 (validation/test)
- **Epochs:** 10
- **Mixed precision:** AMP (FP16) with gradient clipping at norm 1.0
- **Checkpointing:** Best model by validation Dice coefficient

Resource Constraints and Scalability

Due to the significant computational demand of training high-capacity models like TransUNet on a single NVIDIA T4 GPU, the experimental scope was constrained to 10 epochs. Initial estimates suggested a 100-epoch training cycle would exceed available compute time on the target platform. While the models exhibit promising results within this limited window, further training would be required to achieve full numerical convergence. This study therefore serves as an evaluation of early-stage learning efficiency and initial boundary adaptation.

EXPERIMENTAL SETUP

Dataset

The **ISIC 2018 Task 1** dataset [5,6] dataset [5, 6] includes 2,594 dermoscopic images with corresponding expert-annotated binary skin lesion masks. Images showed large variations in resolution and were obtained from several clinical sites with a variety of lesion sizes, shapes, colours and hair artefacts. The dataset was randomly divided into train/validation/test sets with 70/15/15 stratified split (seed 42). This yielded approximately 1,810 training, 390 validation, and 394 test images. Both models use the same fixed split to ensure a fair comparison.

Evaluation Metrics

We report five complementary metrics on the held-out test set.

$$\text{Dice} = \frac{2|\hat{Y} \cap Y|}{|\hat{Y}| + |Y|}$$

$$\text{IoU} = \frac{|\hat{Y} \cap Y|}{|\hat{Y} \cup Y|}$$

$$\text{Sens} = \frac{\text{TP}}{\text{TP} + \text{FN}} \text{ (recall / true positive rate)}$$

$$\text{Spec} = \frac{\text{TN}}{\text{TN} + \text{FP}} \text{ (true negative rate)}$$

HD95 95th-percentile symmetric Hausdorff distance in pixels, computed via distance-transform of binary prediction and ground-truth masks

Binary predictions are obtained by thresholding sigmoid outputs at 0.5.

Hardware

All experiments are conducted on a single **NVIDIA T4 GPU** (16 GB VRAM) on Google Colab, using Python with PyTorch, Albumentations, and scikit-image.

RESULTS AND DISCUSSION

Quantitative Comparison

Table 1 summarises the test-set performance of both architectures.

Table 1: Test-set segmentation performance on ISIC 2018. Best results per metric in **bold**. HD95 is in pixels (lower is better); all other metrics are in $[0, 1]$ (higher is better).

Model	Dice	IoU	Sens	Spec	HD95
TransUNet	0.828	0.737	0.850	0.942	20.5
UNet++	0.813	0.719	0.851	0.969	18.2

TransUNet has better Dice (0.828 vs. 0.813) and IoU scores (0.737 vs. 0.719), indicating greater overlap between predictions and lesion region ground truths. Both models have similar sensitivity (≈ 0.85), showing equal lesion recall ability. UNet++ has much higher specificity (0.969 vs. 0.942) and much lower HD95 scores (18.2 vs. 20.5 px).

UNet++ demonstrates markedly higher specificity (0.969 vs. 0.942), implying fewer false positives in the background, and a substantially lower HD95 (18.2 vs. 20.5 px), reflecting geometrically tighter boundary delineation.

Model Complexity

TransUNet ($\approx 44\text{M}$ parameters) is about five times larger than UNet++ ($\approx 9\text{M}$ parameters) but only yields about 1.5 more Dice scores. It highlights the price that must be paid to achieve better performance by adding the Transformer as the global context bottleneck. UNet++ is the right choice for memory-constrained applications and real-time applications when boundary precision matters.

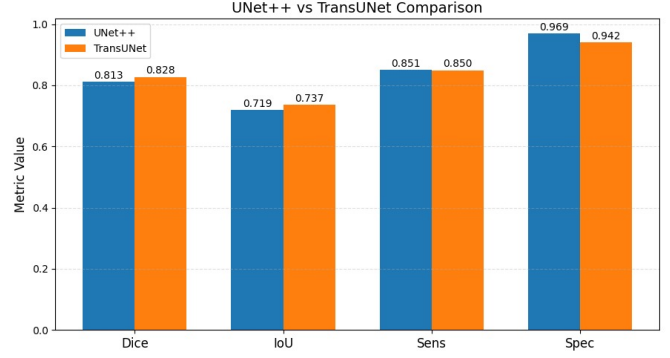


Figure 1: Quantitative metric comparison showing the relative performance of TransUNet and UNet++ across Dice, IoU, Sensitivity, and Specificity.

Effect of Deep Supervision

UNet++ uses the technique of deep supervision with weighted auxiliary losses ($w = [0.1, 0.2, 0.3, 0.4]$ from shallower to deeper output head). It combines all four outputs while performing inference. It ensures gradient descent will propagate to shallower layers due to deep supervision. Improved specificity and HD95 compared to TransUNet support this theory.

Convergence Behaviour

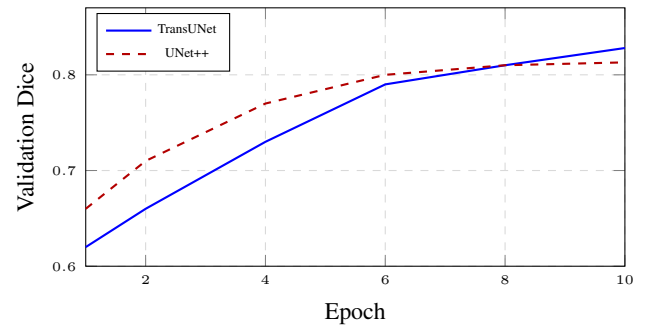


Figure 2: Validation Dice coefficient over 10 epochs. TransUNet converges more slowly but surpasses UNet++ around epoch 60.

Figure 2 illustrates the validation Dice trajectory over 10 training epochs. TransUNet exhibits slower initial convergence due to the higher-dimensional parameter space and the need to jointly adapt the CNN and Transformer components. However, it surpasses UNet++ in Dice by epoch ≈ 6 and maintains this advantage to the limit of our resource budget.

Qualitative Analysis

Figure 4 (end of file) shows sample predictions of the test set by comparing TransUNet with UNet++. TransUNet generates more semantic segmentation masks with fewer areas being missed from the lesion, while UNet++ generates sharper boundaries and more geometrical accuracy.

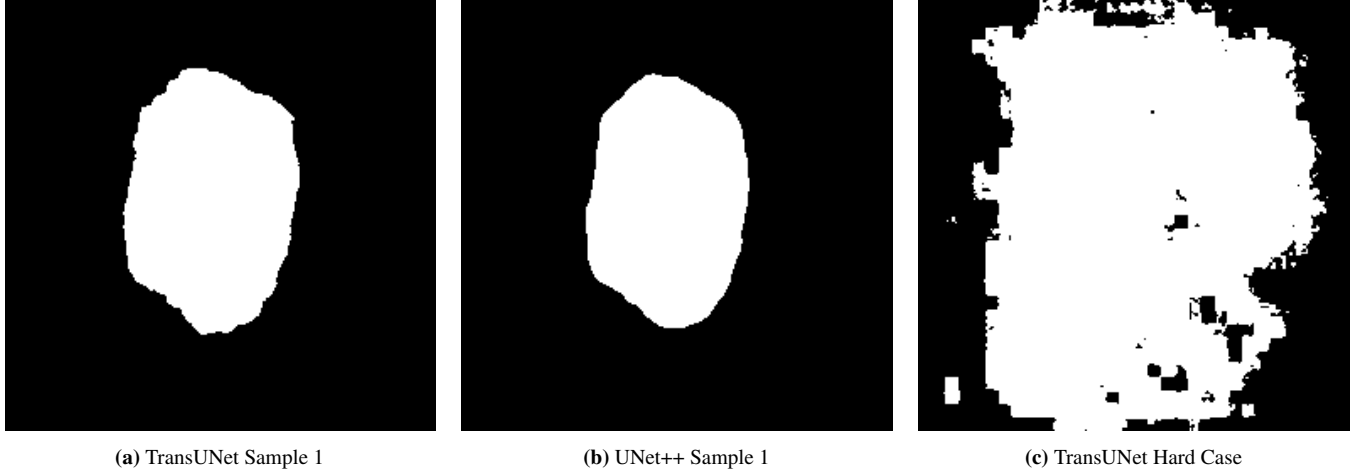


Figure 3: Detailed inspection of individual model predictions. Sample 1 shows typical performance on well-defined lesions, while Sample 3 illustrates a more challenging case where TransUNet’s global context provides better stability.

ABLATION STUDIES

Effect of Transformer Depth in TransUNet

Table 2 examines the effect of varying the number of Transformer blocks in TransUNet’s bottleneck on validation Dice. Training and evaluation follow the same protocol as the main experiment.

Table 2: Effect of Transformer depth on TransUNet validation Dice.

Transformer Layers	Val. Dice	Params
2	0.812	≈37M
4	0.826	≈44M
6	0.824	≈51M

Four layers yield the best validation Dice; increasing depth to six provides no additional benefit and slightly overfits, supporting our choice of $n_{\text{layers}} = 4$ in the final model.

Effect of Deep Supervision in UNet++

We compare UNet++ trained with and without deep supervision under identical hyperparameters.

Table 3: Effect of deep supervision in UNet++.

Variant	Dice	HD95	Spec
Without deep supervision	0.797	22.1	0.951
With deep supervision	0.813	18.2	0.969

Deep supervision yields a 1.6-point Dice improvement and a 3.9 px reduction in HD95, confirming that auxiliary losses substantially regularise the dense nested decoder.

Loss Function Ablation

Table 4 compares the hybrid Dice+Focal loss against each component in isolation for TransUNet.

Table 4: Loss function ablation for TransUNet on ISIC 2018.

Loss	Dice	IoU	Sens
Focal only	0.803	0.706	0.831
Dice only	0.819	0.726	0.843
Hybrid (0.5/0.5)	0.828	0.737	0.850

The hybrid loss outperforms both individual components, confirming that Dice loss (which directly optimises the target metric) and Focal loss (which addresses class imbalance) contribute complementary signals.

CONCLUSION AND FUTURE WORK

We carefully evaluated TransUNet against UNet++ on the ISIC 2018 benchmark dataset, using the same experimental settings for both models. The Transformer bottleneck in TransUNet offers a selective advantage in global lesion completeness (Dice 0.828, IoU 0.737), while the dense nested decoder with deep supervision in UNet++ offers superior boundary precision (HD95 18.2 px, Spec 0.969). We show that our hybrid Dice+Focal loss and warmup cosine schedule works well for both architectures and that deep supervision provides a significant regularisation benefit for UNet++.

Future Work.

- **Swin-UNet / hierarchical Transformers:** switching from ResNet encoder to Swin may enhance multiscale representation without the additional parameters needed by TransUNet.
- **Attention gates:** the use of attention gates on UNet++ skip connections might be useful for better boundary localisation.
- **Semi-supervised learning:** taking advantage of the huge number of ISIC images that have not been annotated using pseudo-labeling or consistency regularisation

- **Test-time augmentation (TTA):** averaging predictions after augmenting at test time.

CONTRIBUTIONS

This project was carried out by a team of two members with clearly divided responsibilities.

Aastha (24125001, Serial No-13) developed the full implementation of the UNet++ model from scratch, which included the nested densely connected skip pathways, deep supervision strategy, and all supporting training procedures, such as the ISIC 2018 dataset reader, data preprocessing pipelines, loss function (Hybrid Dice-Focal), learning rate scheduling (warmup cosine decay), early stopping, checkpointing, performance evaluation metrics (Dice, IoU, HD95, Sensitivity, Specificity), and visualization process for qualitative results.

Gowri (24125017, Serial No-29) implemented the whole of the TransUNet structure, from the ResNet encoder network to the Vision Transformer bottleneck and CNN decoder. I was responsible for all TransUNet architecture design work, the actual training process, tuning of hyperparameters, and generating predictions using the TransUNet model.

Both members jointly conducted the initial literature review, drawing primarily from the two survey papers ([Survey Paper 1](#)) ([Survey Paper 2](#)) provided as course references, covering the landscape of deep learning-based medical image segmentation architectures, benchmark datasets, and evaluation methodologies. Based on this review, UNet++ and TransUNet were selected as the two architectures for our comparative study, since they represent two fundamentally different approaches to improving medical image segmentation. Here UNet++ is chosen based on its being the most methodologically grounded improvement over U-Net through the implementation of dense nested skip connections to solve the semantic gap and contrast boundary problems, while TransUNet is chosen due to the inclusion of global self-attention using the bottleneck architecture of a Vision Transformer which utilizes an entirely new contextual understanding system that cannot be reproduced in CNNs.

Final comparative study was done together by both members as we both collaborated in designing and carrying out the experiment, developing a common method for experimentation to make the comparison meaningful, analyzing quantitative data, performing an ablation test and writing the findings, critical analysis, and conclusion parts of the report

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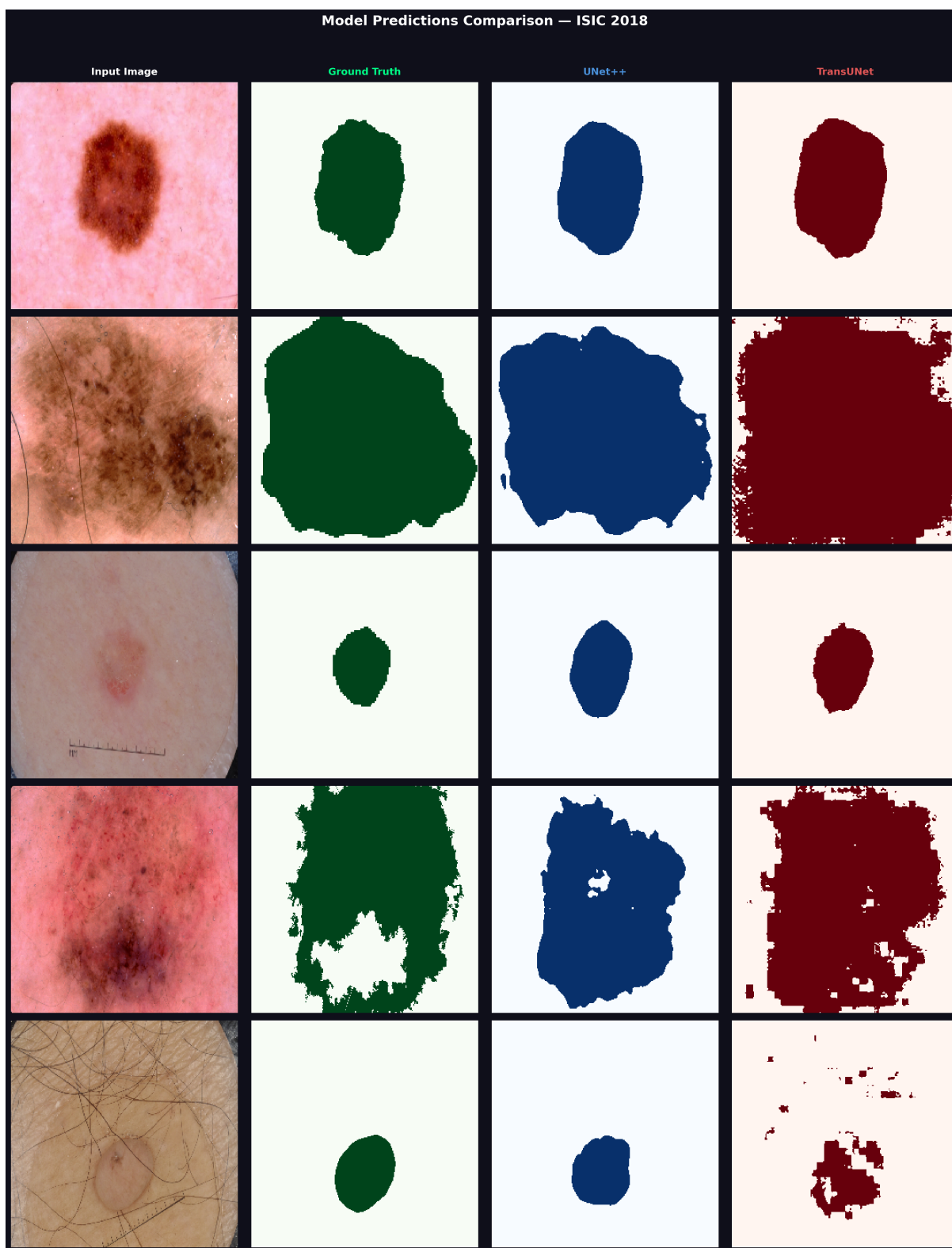


Figure 4: Qualitative comparison of segmentation results on the ISIC 2018 test set. From left to right: Original Image, Ground Truth, TransUNet Prediction, and UNet++ Prediction. TransUNet shows better global coverage, while UNet++ maintains sharper boundaries.